1. **Course number and name**
   CHE 453 Fundamentals of Electrochemistry

2. **Credits, contact hours and categorization**
   Credit hours: 3. Contact hours: 3 lecture hours per week.

3. **Instructor's or course coordinator's name**
   Brian P. Chaplin

4. **Text book, title, author, publisher, year**
   (Required) Bard, A. J.; Faulkner, L. R., *Electrochemical Methods: Fundamentals and Applications*. Second ed.; John Wiley and Sons: 2001; p 833.
   a. Other supplemental materials

      Bockris, J. O. M.; Reddy, A. K.; Gambra-Aldeco, M., *Modern Electrochemistry*. 2nd ed.; Plenum Press: New York, 2000.

      Orazem, M. E.; Tribollet, B., *Electrochemical Impedance Spectroscopy*. John Wiley and Sons: 2008; p 525.

      A. Lasia, Electrochemical Impedance Spectroscopy and Its Applications, In: *Modern Aspects of Electrochemistry*, B. E. Conway, J. Bockris, and R.E. White, Eds., Kluwer Academic/Plenum Publishers, New York, 1999, Vol. 32, p. 143-248.

5. **Specific course information**
   a. Brief description of the content of the course (catalog description)
      Introduction to the fundamentals of electrochemistry and its application in a variety of technologies (i.e., batteries, fuel cells, electrolysis cells). Includes methods for the analysis of cells using electrochemical techniques.
   b. Prerequisites or co-requisites
      Graduate or advanced undergraduate standing.
   c. Indicate whether a required, elective, or selected elective.
      Elective.

6. **Specific goals for the course**
   a. Specific outcomes of instruction.

| | |
|---|---|
| 1 | Use basic Excel cell functionality to tabulate and graph electrochemical data derived from cyclic voltammetry and electrochemical impedance spectroscopy experiments, and interpret with mathematical models |
| 2 | Solving differential equations by the Laplace transform technique, with application to the diffusion equation |
| 3 | Solve partial differential equations numerically and apply to analyze linear sweep and cyclic voltammetry data |
| 4 | Using complex variable notation to derive equivalent circuit model fits of experimental electrochemical impedance spectroscopy data and compare to computer generated fits using Gamry software |
| 5 | Review a published paper in the areas of electrochemistry or electrochemical |

| | |
|---|---|
| | technologies, and prepare a presentation that summarizes and critiques the paper. |

b. Explicitly indicate which of the student outcomes are addressed by the course.

| Student Learning Outcomes | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| (1) an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics. | | | | X |
| (2) an ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors. | X | | | |
| (3) an ability to communicate effectively with a range of audiences. | | | X | |
| (4) an ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts. | X | | | |
| (5) an ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives. | X | | | |
| (6) an ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions. | | | X | |
| (7) an ability to acquire and apply new knowledge as needed, using appropriate learning strategies. | | | | X |

## 7. Brief list of topics to be covered

- Week #1. Introduction and overview of electrochemical cells and electrode processes
- Week #2-3. Thermodynamics of electrochemical cells
- Week #4-6. Kinetics of electrochemical reactions
- Week #7. Mass transfer in electrochemical systems
- Week #8. Adsorption and double-layer structure
- Week #9-11. Electrochemical methods including potential step and sweep methods and controlled current techniques
- Week #12-13. Numerical simulations applied to potential step and sweep methods
- Week #14. Electrochemical impedance spectroscopy of electrochemical cells
- Week #15. Student presentations